

Statistics Lecture 6

Hypothesis testing pt. 2

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AALBORG UNIVERSITY
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Connectivity

Schedule

Lecture

1. Introduction to statistics
2. Parameter estimation
3. Confidence intervals
4. Regression
5. Hypothesis testing pt. 1

6. Hypothesis testing pt. 2

Workshop: Wrap-up and exam problems

When

March 12

March 24

March 26

March 31

April 7

April 9

April 28?

Outline

Recap on hypothesis testing part 1

Advanced hypothesis testing

1. Tests with one normally distributed population and unknown variance
2. Tests with two populations
3. Tests on the slope

Recap on hypothesis testing

How do we conduct the tests?

We need to formulate two opposing hypotheses:

H_0 : The **null hypothesis**, the one assumed to be true

H_1 : The **alternative hypothesis**, which contradicts H_0

We try to find evidence to support H_0

If we cannot, then we say we can **reject** H_0

Accepting a hypothesis does not mean it is true, but that the data support it

Procedure for testing

1. Identify the underlying distribution F
2. Choose a parameter for testing θ
3. Formulate the null hypothesis H_0 and the alternative hypothesis H_1 about θ
4. Design the test (test statistic T or p-value ν) and their formulation
5. Define the rejection region R
6. Collect a sample $X_1, X_2, X_3, \dots, X_n$ of i.i.d. RVs
7. Calculate the test statistic T (and p-value if necessary)
8. Decide:
 - **If** the observation regarding θ is close enough to the value(s) in H_0 : Accept H_0
 - **Else**: reject H_0 and accept H_1

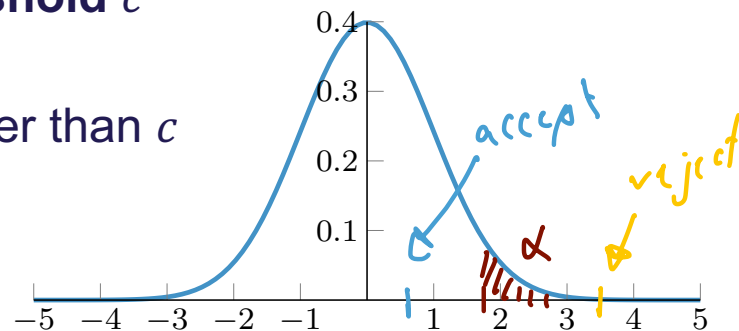
Two ways of testing

$$H_0: \theta \leq \theta_0$$
$$c = z_\alpha$$

1. Comparing the test statistic T with the threshold c

These values are points in the x-axis

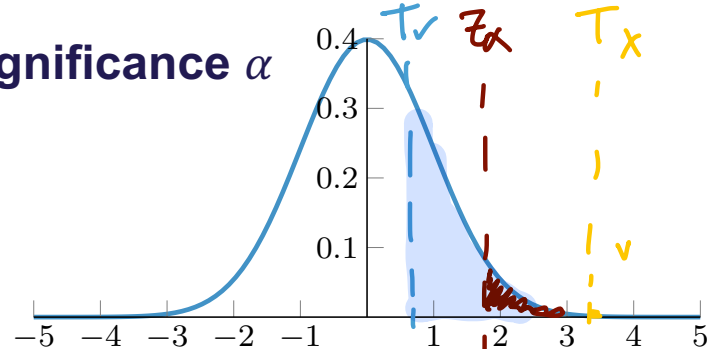
Reject the null hypothesis if T is farther from center than c



2. Comparing the p-value v with the level of significance α

These are the areas under the curve

Reject the null hypothesis if $v < \alpha$



Summary from Hypothesis testing pt. 1

Two-sided test: Null hypothesis has the form $H_0: \theta = \theta_0$

One-sided test: Null hypothesis has the form $H_0: \theta \leq \theta_0$ or $\theta \geq \theta_0$

Use Z-tests when population is normally distributed and σ is known

We know that $F \sim N(\mu, \sigma^2)$, so we use the quantiles Z_α or $Z_{\alpha/2}$

Why? A sum of n normal RVs always has a normal distribution $N(n\mu, n\sigma^2)$

Note: Same as in Confidence Intervals

Binomial test (one-sided) when population is Bernoulli distributed

We know that $F \sim \text{Bernoulli}(p)$, so we calculate the p-value as the CDF of Binomial RVs

Why? A sum of n Bernoulli RVs always has Binomial distribution $B(n, p)$

Tests for the mean:
Normal population with unknown variance

The t-test

If σ^2 is known:

We can use $N(0,1)$ and its quantiles Z_α and $Z_{\alpha/2}$

If σ^2 is **unknown**:

It must be estimated and we cannot use $N(0,1)$ anymore

We estimate σ^2 using our unbiased estimator for the sample variance

$$\hat{\sigma}_n^2 = S_n^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2$$

And define the **test statistic** based on S_n^2

T-test: 

Since we are not using σ^2 , we now have to use the **t-distribution** for testing

Two-sided test for μ with normal population and **unknown** σ

The null hypothesis $H_0: \mu = \mu_0$, so $H_1: \mu \neq \mu_0$

Test statistic: $T = |\hat{\mu}_n - \mu_0| \sqrt{n} / S_n$

Rejection region: $R = \{X_1, X_2, \dots, X_n : \bar{T} > c\}$

The total area of the region outside the CI is α

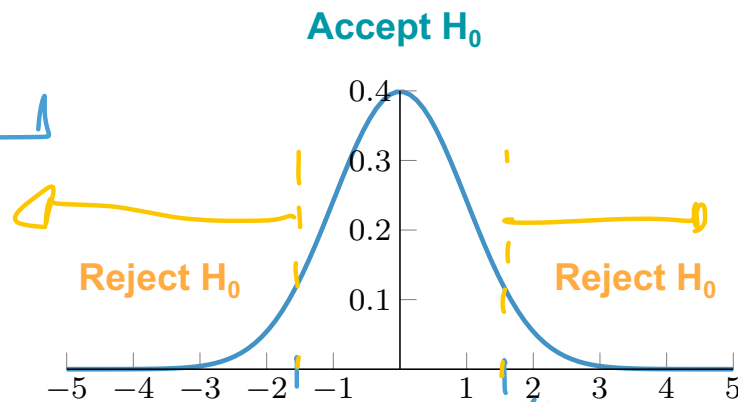
$$C_{1-\alpha} = \left(\hat{\mu}_n - T_{n-1, \alpha/2} \frac{S_n}{\sqrt{n}}, \hat{\mu}_n + T_{n-1, \alpha/2} \frac{S_n}{\sqrt{n}} \right)$$

So $c = T_{n-1, \alpha/2}$

Reject H_0 if $\frac{|\hat{\mu}_n - \mu_0| \sqrt{n}}{S_n} > T_{n-1, \alpha/2}$

estimate for std. deviation

use t-distribution instead of std. normal



Example

A group of 50 patients with high cholesterol levels were given a new drug. The reduction in levels of cholesterol were recorded.

The reduction of cholesterol levels had a sample mean of $\hat{\mu}_n = 14.8$ with sample standard deviation of 6.4. Was the drug effective with 5% significance?

1. Formulate the null and alternative hypothesis

$$H_0: \mu = \mu_0 = 0 \text{ and } H_1: \mu \neq \mu_0$$

2. Calculate the test statistic

$$T = |\hat{\mu}_n - \mu_0| \sqrt{n} / S_n = |14.8 - 0| \sqrt{50} / 6.4 = 16.35$$

3. Can you reject the null hypothesis?

$$\text{Reject if } T > T_{n-1, \alpha/2}$$

$$z_{\alpha/2} = 1.96$$

Solution

A group of 50 patients with high cholesterol levels were given a new drug. The reduction in levels of cholesterol were recorded.

The reduction of cholesterol levels had a sample mean of 14.8 with sample standard deviation of 6.4. Was the drug effective with 5% significance?

1. Formulate the null and alternative hypothesis

$$H_0: \mu = 0 \text{ and } H_1: \mu \neq 0$$

2. Calculate the test statistic

$$T = |\hat{\mu}_n - \mu_0| \sqrt{n} / S_n = |14.8 - 0| \sqrt{50} / 6.4 = 16.35$$

3. Can you reject the null hypothesis?

Reject if $T > T_{n-1, \alpha/2}$. Since $T = 16.35 > 2.009$, we reject H_0

One-sided t-test for μ with normal population and **unknown** σ

The null hypothesis $H_0: \mu \leq \mu_0$

$H_1: \mu > \mu_0$

Test statistic: $T = (\hat{\mu}_n - \mu_0)\sqrt{n}/S_n$

Rejection region:

$$R = \{X_1, X_2, \dots, X_n : T > T_{n-1, \alpha}\}$$

$$\text{CI is } C_{1-\alpha} = \left(-\infty, \hat{\mu}_n + T_{n-1, \alpha} \frac{S_n}{\sqrt{n}}\right)$$

Reject H_0 if $\frac{(\hat{\mu}_n - \mu_0)\sqrt{n}}{S_n} > T_{n-1, \alpha}$

$T =$

The null hypothesis $H_0: \mu \geq \mu_0$

$H_1: \mu < \mu_0$

Test statistic: $T = (\hat{\mu}_n - \mu_0)\sqrt{n}/S_n$

Rejection region:

$$R = \{X_1, X_2, \dots, X_n : T < -T_{n-1, \alpha}\}$$

$$\text{CI is } C_{1-\alpha} = \left(\hat{\mu}_n - T_{n-1, \alpha} \frac{S_n}{\sqrt{n}}, \infty\right)$$

Reject H_0 if $\frac{(\hat{\mu}_n - \mu_0)\sqrt{n}}{S_n} < -T_{n-1, \alpha}$

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Summary table

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Testing for the mean of one normal population with **unknown** variance

H_0	H_1	Test statistic T	Rejection region	P-value
$\mu = \mu_0$	$\mu \neq \mu_0$	$ \hat{\mu}_n - \mu_0 \sqrt{n}/S_n$	$T > T_{n-1, \alpha/2}$	$2(1 - P(T_{n-1} \leq T))$
$\mu \leq \mu_0$	$\mu > \mu_0$	$(\hat{\mu}_n - \mu_0) \sqrt{n}/S_n$	$T > T_{n-1, \alpha}$	$(1 - P(T_{n-1} \leq T))$
$\mu \geq \mu_0$	$\mu < \mu_0$	$(\hat{\mu}_n - \mu_0) \sqrt{n}/S_n$	$T < -T_{n-1, \alpha}$	$P(T_{n-1} \leq T)$

Using test statistic

Using p-value

Syntax table

Parameter	Matlab syntax	Python syntax
<i>quantile</i> → $T_{n-1, \alpha/2}$	<code>tinvs(1-α/2, n-1)</code>	<code>scipy.stats.t(n-1).ppf(1-α/2)</code>
<i>CDF of t-dist.</i> → $P(T_{n-1} \leq T)$	<code>tcdf(T, n-1)</code>	<code>scipy.stats.t(n-1).cdf(T)</code>

Tests with two normal populations: Testing the equality of means

Two-sided test with known variance

We want to test $\bar{X} - \bar{Y}$ where both RVs have normal distribution

Null hypothesis is H_0 : $\mu_X = \mu_Y$ and so $\mu_X - \mu_Y = 0$

We calculate $\mu_{\bar{X}-\bar{Y}} = \mu_X - \mu_Y$ $\text{var}(\bar{X} - \bar{Y}) = \frac{\sigma_X^2}{n_X} + \frac{\sigma_Y^2}{n_Y}$

Test statistic: $T = \frac{|\hat{\mu}_X - \hat{\mu}_Y|}{\sqrt{\frac{\sigma_X^2}{n_X} + \frac{\sigma_Y^2}{n_Y}}}$

Rejection region: $R = \{X_1, X_2, \dots, X_n : T > Z_{\alpha/2}\}$

Reject H_0 if $\frac{|\hat{\mu}_X - \hat{\mu}_Y|}{\sqrt{\frac{\sigma_X^2}{n_X} + \frac{\sigma_Y^2}{n_Y}}} > Z_{\alpha/2}$

α : significance level

$$\text{var}(X) = \sigma_X^2$$

$$\text{var}(Y) = \sigma_Y^2$$

Two-sided test with **unknown** variance

Null hypothesis is $H_0: \mu_X = \mu_Y$ and

We calculate $\mu_{\bar{X}-\bar{Y}} = \mu_X - \mu_Y$

Non-biased estimator for the variance is $S_{\bar{X}-\bar{Y}}^2 = \frac{\sum_{i=1}^{n_X} (X_i - \bar{X})^2 + \sum_{j=1}^{n_Y} (Y_j - \bar{Y})^2}{n_X + n_Y - 2}$

estimate

Test statistic: $T = \frac{|\hat{\mu}_X - \hat{\mu}_Y|}{S_{\bar{X}-\bar{Y}} \sqrt{\frac{1}{n_X} + \frac{1}{n_Y}}}$

Rejection region: $R = \{X_1, X_2, \dots, X_n : T > T_{n_X+n_Y-2, \alpha/2}\}$

Reject H_0 if $\frac{|\hat{\mu}_X - \hat{\mu}_Y|}{S_{\bar{X}-\bar{Y}} \sqrt{\frac{1}{n_X} + \frac{1}{n_Y}}} > T_{n_X+n_Y-2, \alpha/2}$

T =

$n_X - 1$ for X
 $n_Y - 1$ for Y

Exercise

A group of 22 volunteers at a cold research institute caught a cold.
A random selection of 10 of these were given tablets with vitamin C
Others were given a placebo and the time that the cold lasted was recorded
Did the vitamin C have an effect on the mean length of the cold?

At what level of significance?

Null hypothesis:

$$H_0: \mu_X = \mu_Y$$

$$\mu_X - \mu_Y = 0$$

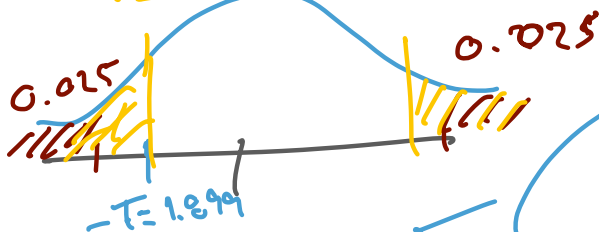
$$n_X = 10 \quad n_Y = 12$$

X		Y	
Treated with vitamin C		Given Placebo	
5.5	6.0	6.5	7.5
6.0	7.5	6.0	6.5
7.0	5.5	8.5	7.5
6.0	7.0	7.0	6.0
7.5	6.5	6.5	8.5
		8.0	7.0

Exercise

We calculate

$$v = 0.0721$$



$$-T_{n_X + n_Y - 2, \alpha/2} = -2.0859$$

val. C

placebo

$$\hat{\mu}_X = 6.450 \text{ and } \hat{\mu}_Y = 7.125$$

$$S_{\bar{X} - \bar{Y}}^2 = 0.689$$

$$T = \frac{|\hat{\mu}_X - \hat{\mu}_Y|}{S_{\bar{X} - \bar{Y}} \sqrt{\frac{1}{n_X} + \frac{1}{n_Y}}} = 1.899$$

quantile of t-dist.

$$\hat{\mu}_X - \hat{\mu}_Y = -0.675$$

Since $T = 1.899 < T_{n_X + n_Y - 2, \alpha/2} = 2.0859$, we cannot reject H_0 with $\alpha = 0.05$

The p-value is $v = 0.0721$, so we cannot reject H_0 for all $\alpha < 0.0721$

Tests using regression

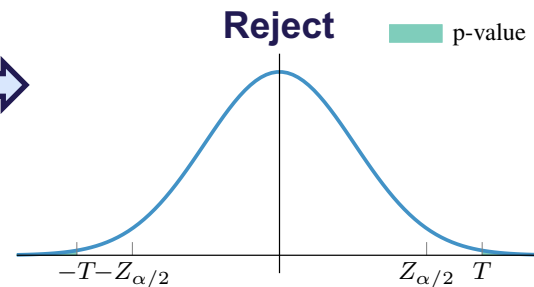
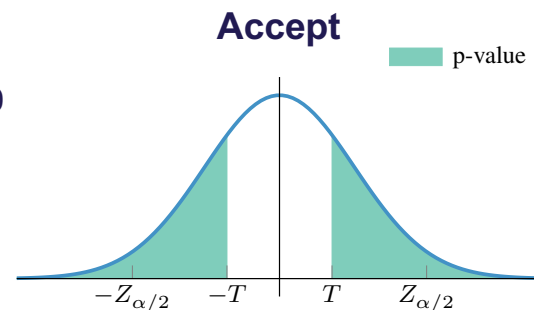
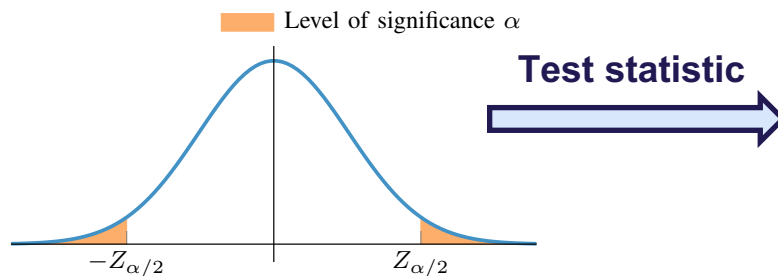
The limits of hypothesis testing

H_0 : The **null hypothesis**, the one assumed to be true

H_1 : The **alternative hypothesis**, which contradicts H_0

We try to find evidence to reject H_0

But we don't want to reject H_0 when it's true: α



What if we want to test tendencies in the data?

Example

A guy claims that how fast one drives has no impact on fuel consumption of a car
To test this, measurements were collected for a car driving at different speeds
Can we reject the claim with these data?

Speed (mph)	Miles per gallon
45	24.2
50	25.0
55	23.3
60	22.0
65	21.5
70	20.6
75	19.8

Inference on the slope

fuel consumption $\rightarrow$ speed

$$Y_i = \beta_0 + X_i \beta_1$$

Does X_i have an effect on Y_i ?

If $\beta_1 = 0$, there is no effect

But we don't have β_1 , we're stuck with its estimator $\hat{\beta}_1$

We can do hypothesis testing $H_0: \beta_1 = \beta_H = 0$

Due to asymptotic normality

$$\hat{\beta}_1 \sim N\left(\beta_1, \frac{\sigma^2}{S_X^2 n}\right), \text{ then } \frac{(\hat{\beta}_1 - \beta_1) \times S_X \sqrt{n}}{\sigma} \sim N(0,1)$$

Hypothesis testing

We use $S_X^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2$ to calculate

biased
version

$$\underline{\widehat{\text{se}}(\hat{\beta}_1)} = \frac{\hat{\sigma}}{S_X \sqrt{n}} = \frac{\hat{\sigma}}{\sqrt{\sum_{i=1}^n (X_i - \bar{X}_n)^2}}$$

To get

$$\boxed{\frac{\hat{\beta}_1 - \beta_1}{\widehat{\text{se}}(\hat{\beta}_1)}} = \frac{(\hat{\beta}_1 - \beta_1) S_X \sqrt{n}}{\hat{\sigma}} = \frac{(\hat{\beta}_1 - \beta_1) S_X \sqrt{n}}{\sqrt{\left(\frac{1}{n-2}\right) \sum_{i=1}^n \hat{\epsilon}_i^2}} \sim \underline{T_{n-2}}$$

Degrees of freedom: no. of independent variables minus the no. of equations

We have n values and one equation for each β_0 and β_1 , so $n - 2$ dofs

Estimated standard errors

We already had an unbiased estimator

$$\hat{\sigma}^2 = \left(\frac{1}{n-2}\right) \sum_{i=1}^n \hat{\epsilon}_i^2 = \left(\frac{1}{n-2}\right) \sum_{i=1}^n \left(Y_i - (\hat{\beta}_0 + \hat{\beta}_1 X_i)\right)^2$$

$$\text{var}(\hat{\beta}_0) = \frac{\hat{\sigma}^2}{S_X^2 n} \left(\frac{\sum_{i=1}^n X_i^2}{n} \right)$$

$$\text{var}(\hat{\beta}_1) = \frac{\hat{\sigma}^2}{S_X^2 n}$$

$$\widehat{\text{se}}(\hat{\beta}_0) = \frac{\hat{\sigma}}{S_X \sqrt{n}} \sqrt{\frac{\sum_{i=1}^n X_i^2}{n}}$$

$$\widehat{\text{se}}(\hat{\beta}_1) = \frac{\hat{\sigma}}{S_X \sqrt{n}}$$

Properties of the estimators

Under appropriate conditions we have

1. Consistency: $\hat{\beta}_0 \xrightarrow{P} \beta_0$ and $\hat{\beta}_1 \xrightarrow{P} \beta_1$

2. Asymptotic Normality:

Both $\frac{\hat{\beta}_0 - \beta_0}{\widehat{\text{se}}(\hat{\beta}_0)}$ and $\frac{\hat{\beta}_1 - \beta_1}{\widehat{\text{se}}(\hat{\beta}_1)}$ are asymptotically standard Normal RVs: $N(0,1)$

3. Approximate $1 - \alpha$ confidence intervals for β_0 and β_1

$$\hat{\beta}_0 \pm Z_{\alpha/2} \widehat{\text{se}}(\hat{\beta}_0) \text{ and } \hat{\beta}_1 \pm Z_{\alpha/2} \widehat{\text{se}}(\hat{\beta}_1)$$

Test statistic and rejection region

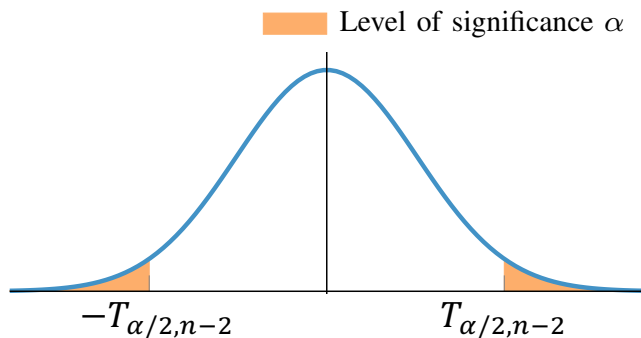
Test statistic for two-sided test is

$$T = \frac{|\hat{\beta}_1 - \beta_H|}{\widehat{\text{se}}(\hat{\beta}_1)} = \frac{|\hat{\beta}_1 - \beta_H| S_X \sqrt{n}}{\left(\frac{1}{n-2}\right) \sum_{i=1}^n \hat{\epsilon}_i^2}$$

std. error (pointing to $\widehat{\text{se}}(\hat{\beta}_1)$)
residuals (pointing to $\sum_{i=1}^n \hat{\epsilon}_i^2$)

Rejection region for level of significance α is $R = \{\hat{\beta}_1 : T > \underline{T_{n-2, \alpha/2}}\}$

P-value: same as before



Solving the example

$$H_0: \beta_1 = \beta_H = 0$$

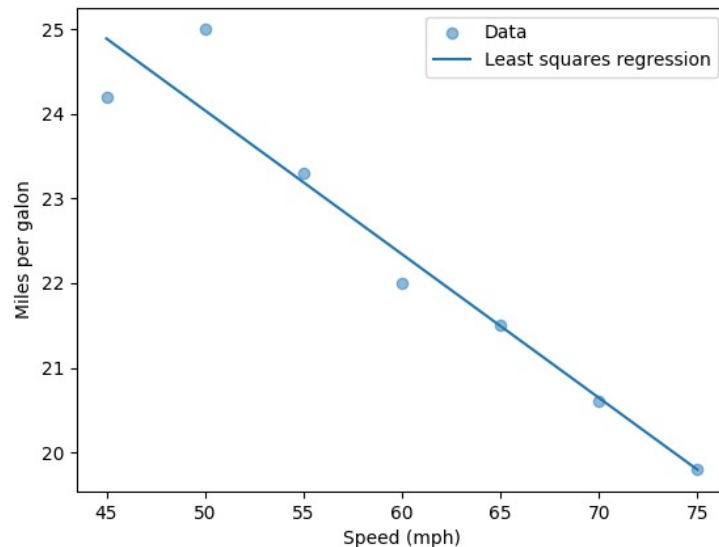
$$n = 7$$

Estimators: $\hat{\beta}_0 = 32.542$ and $\hat{\beta}_1 = -0.169$ and $\widehat{se}(\hat{\beta}_1) = 0.0208$

$$T = \frac{|\hat{\beta}_1 - 0|}{\widehat{se}(\hat{\beta}_1)} = 8.139 > 2.5705: \text{reject } H_0$$

$$t_{n-2, \alpha/2}$$

Speed (mph)	Miles per gallon
45	24.2
50	25.0
55	23.3
60	22.0
65	21.5
70	20.6
75	19.8



Summary

Summary

The type of test to use depends on:

Number of populations involved, types of hypothesis, and assumptions

- **Z-test:** for normal population and the variance is **known**
- **Binomial test:** When population is Bernoulli
- **T-test:** for normal population and the variance is **unknown**

We need to adapt the estimator for the variance when we have two populations

But, if the sample size is large, we can use the **Central Limit Theorem**